Table 1: Moment of inertia of the disk about its central axis

Quantity	Value
α (rad s^{-2})	0.5590 ± 0.0017
α_f (rad s^{-2})	-0.05986 ± 0.00058
$I_{\text{hori,obs}}$ (kg m^2)	$(9.713 \pm 0.028) \times 10^{-3}$
Disk mass M (g)	1445.3
Disk diameter $2R$ (cm)	22.84
$I_{\text{hori,ref}} = MR^2/2$ (kg m^2)	0.009425
Error	3.06%

Table 2: Moment of inertia of the disk about its diameter axis

Quantity	Value
α (rad s^{-2})	1.0747 ± 0.0018
α_f (rad s^{-2})	-0.14380 ± 0.00052
$I_{\text{verti,obs}}$ (kg m^2)	$(4.9293 \pm 0.0074) \times 10^{-3}$
Disk thickness L (mm)	25.40
$I_{\text{verti,ref}} = MR^2/4 + ML^2/12$ (kg m^2)	0.004790
Error	2.91%

Table 3: Moment of inertia of the whole system with the disk offset by $d = 10.00\,\text{cm}$

Quantity	Value
α (rad s^{-2})	0.086260 ± 0.000058
α_f (rad s^{-2})	-0.02299 ± 0.00021
I_{off} (kg m^2)	0.05494 ± 0.00011

Table 4: Moment of inertia of the whole system with the disk offset by $d = 11.00\,\mathrm{cm}$

Quantity	Value
α (rad s^{-2})	0.082854 ± 0.000060
α_f (rad s^{-2})	-0.018778 ± 0.000098
I_{off} (kg m^2)	0.059055 ± 0.000067

Table 5: Moment of inertia of the whole system with the disk offset by $d = 12.00\,\mathrm{cm}$

Quantity	Value
α (rad s^{-2})	0.074822 ± 0.000044
α_f (rad s^{-2})	-0.020235 ± 0.000051
I_{off} (kg m^2)	0.063269 ± 0.000045

Table 6: Moment of inertia of the whole system with the disk offset by $d = 13.00\,\mathrm{cm}$

Quantity	Value
α (rad s^{-2})	0.071605 ± 0.000056
α_f (rad s^{-2})	-0.02092 ± 0.00013
I_{off} (kg m^2)	0.065002 ± 0.000100

Table 7: Moment of inertia of the whole system with the disk offset by $d = 14.00\,\mathrm{cm}$

Quantity	Value
α (rad s^{-2})	0.064639 ± 0.000036
α_f (rad s^{-2})	-0.02128 ± 0.00017
I_{off} (kg m^2)	0.07000 ± 0.00014

Table 8: Moment of inertia of the rotating platform and counterweight alone (no disk)

Quantity	Value
α (rad s^{-2})	0.15693 ± 0.00016
α_f (rad s^{-2})	-0.03173 ± 0.00037
I_{plat} (kg m^2)	0.031875 ± 0.000067

Table 9: Moment of inertia of the whole system I_{off} at five different disk offsets d

Offset d (cm)	I_{off} (kg m^2)
10.00	0.05494 ± 0.00011
11.00	0.059055 ± 0.000067
12.00	0.063269 ± 0.000045
13.00	0.065002 ± 0.000100
14.00	0.07000 ± 0.00014

Table 10: Parallel axis theorem – comparing the I_{off} - d^2 fit's slope and intercept with theory

Quantity	Value
Slope s (g)	$(1.505 \pm 0.015) \times 10^3$
Disk + adapter mass M_{ref} (g)	1475.7
Error	2.01%
y-intercept b (kg m^2)	0.04099 ± 0.00021
$b - I_{\text{plat}}$ (kg m^2)	$(9.12 \pm 0.22) \times 10^{-3}$
$I_{\text{hori,ref}}$ (kg m^2)	0.009425
Error	-3.28%

Table 11: Torsion constant κ of the suspension wire, calibrated with a disk of known moment of inertia

Quantity	Value
$30T$ (s)	109.880 ± 0.016
Period T (s)	3.66267 ± 0.00051
Calibration disk mass M (g)	660.6
Calibration disk diameter $2R$ (mm)	120.30
Calibration disk's $I = MR^2/2$ (kg m^2)	0.001195
κ (N m rad^{-1})	$(3.51678 \pm 0.00098) \times 10^{-3}$

Table 12: Moment of inertia of the black box about the axis through I_{xx}'s suspension hole

Quantity	Value
$30T$ (s)	119.50 ± 0.11
Period T (s)	3.9833 ± 0.0037
I_{xx} ($\mathrm{kg\,m^2}$)	$(1.4134 \pm 0.0027) \times 10^{-3}$

Table 13: Moment of inertia of the black box about the axis through I_{yy}'s suspension hole

Quantity	Value
$30T$ (s)	118.100 ± 0.095
Period T (s)	3.9367 ± 0.0032
I_{yy} ($\mathrm{kg\,m^2}$)	$(1.3805 \pm 0.0023) \times 10^{-3}$

Table 14: Moment of inertia of the black box about the axis through I_{zz}'s suspension hole

Quantity	Value
$30T$ (s)	117.86 ± 0.14
Period T (s)	3.9287 ± 0.0047
I_{zz} ($\mathrm{kg\,m^2}$)	$(1.3749 \pm 0.0033) \times 10^{-3}$

Table 15: Raw stopwatch readings – Horizontal disk (about its central axis), loaded run

Stopwatch readings T (ms)
7323, 7564, 7794, 8012, 8222, 8423, 8619, 8807, 8991, 9169
9342, 9511, 9675, 9836, 9993, 10147, 10297, 10446, 10591, 10734
10874, 11012

Table 16: Raw stopwatch readings – Horizontal disk (about its central axis), friction (unloaded) run

Stopwatch readings T (ms)
4046, 4185, 4325, 4464, 4604, 4745, 4885, 5025, 5166, 5308
5450, 5591, 5734, 5877, 6019, 6163, 6306, 6450, 6594, 6737
6882, 7027, 7172, 7318, 7463, 7610, 7756, 7903, 8049, 8196
8344, 8491, 8640, 8788, 8938, 9086, 9236, 9385, 9536, 9686
9837, 9987, 10139, 10292, 10443, 10596, 10748, 10901, 11054, 11207
11362, 11516, 11671, 11826, 11981, 12137, 12292, 12449, 12606, 12763
12920, 13078

Table 17: Raw stopwatch readings – Vertical disk (about its diameter axis), loaded run

Stopwatch readings T (ms)
6151, 6421, 6669, 6890, 7094, 7284, 7465, 7636, 7799, 7954
8105, 8248, 8387, 8523, 8654, 8781, 8905, 9026, 9144, 9260
9372, 9483, 9592, 9698, 9803, 9905, 10006, 10106, 10204, 10300
10395

Table 18: Raw stopwatch readings – Vertical disk (about its diameter axis), friction (unloaded) run

Stopwatch readings T (ms)
2155, 2287, 2419, 2551, 2685, 2819, 2954, 3088, 3224, 3360
3496, 3634, 3772, 3911, 4051, 4192, 4332, 4473, 4616, 4759
4902, 5047, 5192, 5338, 5484, 5632, 5779, 5927, 6076, 6226
6376, 6528, 6680, 6834, 6987, 7143, 7299, 7455, 7612, 7771
7929, 8089, 8251, 8414, 8577, 8741, 8907, 9073, 9240, 9408
9577, 9748, 9920, 10093, 10268, 10443, 10620, 10798, 10976, 11156
11336, 11518

Table 19: Raw stopwatch readings – Off-axis disk at $d = 10.00$ cm, loaded run

Stopwatch readings T (ms)
7517, 8385, 9168, 9887, 10559, 11192, 11793, 12368, 12920, 13452
13962, 14455, 14931, 15394, 15846, 16286, 16716, 17136, 17547, 17949
18342, 18728

Table 20: Raw stopwatch readings – Off-axis disk at $d = 10.00$ cm, friction (unloaded) run

Stopwatch readings T (ms)
2338, 2537, 2736, 2935, 3134, 3334, 3534, 3734, 3934, 4136
4336, 4539, 4741, 4943, 5145, 5348, 5551, 5754, 5956, 6161
6365, 6570, 6776, 6980, 7186, 7392, 7597, 7804, 8011, 8217
8425, 8633, 8842, 9050, 9259, 9468, 9677, 9886, 10096, 10306
10517, 10729, 10941, 11152, 11364, 11577, 11789, 12002, 12215, 12430
12644, 12859, 13074, 13289, 13505, 13721, 13937, 14155, 14371, 14589
14807, 15026

Table 21: Raw stopwatch readings – Off-axis disk at $d = 11.00$ cm, loaded run

Stopwatch readings T (ms)
811, 1644, 2405, 3109, 3772, 4402, 5003, 5580, 6133, 6665
7177, 7670, 8146, 8609, 9060, 9499, 9929, 10349, 10762, 11163
11558, 11944

Table 22: Raw stopwatch readings – Off-axis disk at $d = 11.00$ cm, friction (unloaded) run

Stopwatch readings T (ms)
9829, 10080, 10331, 10583, 10836, 11088, 11342, 11596, 11851, 12106
12363, 12619, 12876, 13133, 13390, 13648, 13906, 14165, 14426, 14686
14949, 15210, 15472, 15734, 15997, 16260, 16525, 16789, 17054, 17320
17586, 17854, 18121, 18389, 18658, 18926, 19195, 19465, 19737, 20008
20281, 20554, 20827, 21101, 21375, 21650, 21925, 22202, 22479, 22757
23035, 23315, 23594, 23874, 24154, 24436, 24718, 25000, 25285, 25569
25854, 26140

Table 23: Raw stopwatch readings – Off-axis disk at $d = 12.00$ cm, loaded run

Stopwatch readings T (ms)
1372, 2329, 3208, 4021, 4778, 5493, 6162, 6793, 7393, 7970
8524, 9059, 9579, 10084, 10575, 11053, 11518, 11968, 12409, 12838
13259, 13670

Table 24: Raw stopwatch readings – Off-axis disk at $d = 12.00$ cm, friction (unloaded) run

Stopwatch readings T (ms)
3680, 3980, 4280, 4582, 4884, 5187, 5492, 5797, 6104, 6410
6717, 7026, 7335, 7645, 7956, 8269, 8583, 8897, 9214, 9530
9848, 10165, 10485, 10804, 11127, 11449, 11774, 12098, 12425, 12752
13080, 13409, 13739, 14070, 14402, 14737, 15073, 15410, 15749, 16088
16428, 16768, 17110, 17454, 17799, 18147, 18496, 18847, 19199, 19551
19904, 20259, 20615, 20972, 21332, 21693, 22058, 22423, 22790, 23157
23525, 23895

Table 25: Raw stopwatch readings – Off-axis disk at $d = 13.00$ cm, loaded run

Stopwatch readings T (ms)
2082, 2898, 3652, 4360, 5029, 5667, 6281, 6872, 7442, 7990
8521, 9033, 9529, 10013, 10483, 10942, 11392, 11832, 12264, 12687
13102, 13508

Table 26: Raw stopwatch readings – Off-axis disk at $d = 13.00$ cm, friction (unloaded) run

Stopwatch readings T (ms)
4411, 4640, 4869, 5099, 5328, 5559, 5789, 6021, 6254, 6486
6720, 6953, 7187, 7420, 7655, 7889, 8125, 8361, 8598, 8835
9074, 9312, 9550, 9789, 10027, 10267, 10507, 10747, 10989, 11232
11474, 11716, 11960, 12202, 12446, 12691, 12935, 13181, 13427, 13675
13922, 14169, 14417, 14665, 14914, 15163, 15412, 15663, 15914, 16166
16418, 16671, 16924, 17177, 17430, 17684, 17939, 18194, 18450, 18707
18965, 19222

Table 27: Raw stopwatch readings – Off-axis disk at $d = 14.00$ cm, loaded run

Stopwatch readings T (ms)
6298, 7313, 8245, 9109, 9911, 10662, 11370, 12045, 12686, 13301
13892, 14465, 15023, 15565, 16094, 16606, 17106, 17592, 18066, 18528
18979, 19422

Table 28: Raw stopwatch readings – Off-axis disk at $d = 14.00$ cm, friction (unloaded) run

Stopwatch readings T (ms)
5906, 6119, 6332, 6546, 6760, 6975, 7190, 7406, 7621, 7837
8054, 8270, 8487, 8704, 8922, 9140, 9359, 9578, 9797, 10017
10237, 10457, 10677, 10897, 11119, 11341, 11564, 11787, 12009, 12232
12457, 12680, 12904, 13128, 13354, 13579, 13805, 14032, 14259, 14486
14713, 14941, 15168, 15396, 15626, 15855, 16087, 16317, 16547, 16779
17010, 17241, 17473, 17706, 17939, 18173, 18407, 18642, 18877, 19113
19348, 19584

Table 29: Raw stopwatch readings – Rotating platform + counterweight only (no disk), loaded run

Stopwatch readings T (ms)
7952, 8560, 9116, 9631, 10115, 10577, 11017, 11437, 11838, 12224
12596, 12956, 13305, 13644, 13974, 14295, 14608, 14913, 15211, 15502
15788, 16067

Table 30: Raw stopwatch readings – Rotating platform + counterweight only (no disk), friction (unloaded) run

Stopwatch readings T (ms)
4789, 4954, 5119, 5284, 5450, 5616, 5782, 5947, 6113, 6280
6446, 6613, 6780, 6948, 7116, 7284, 7453, 7621, 7790, 7959
8128, 8296, 8465, 8636, 8806, 8977, 9148, 9319, 9490, 9661
9833, 10004, 10177, 10350, 10523, 10696, 10870, 11043, 11217, 11391
11565, 11739, 11914, 12089, 12265, 12442, 12618, 12794, 12970, 13147
13324, 13500, 13679, 13857, 14035, 14214, 14393, 14572, 14752, 14931
15110, 15290

Table 31: Raw timing measurements for the calibration disk and the three black-box suspension holes

Configuration	T_1 (s)	T_2 (s)	$30T = T_2 - T_1$ (s)
disk	2.21	112.06	109.85
disk	7.95	117.85	109.90
disk	8.15	118.04	109.89
hole1	21.48	141.07	119.59
hole1	12.45	131.73	119.28
hole1	9.47	129.10	119.63
hole2	14.81	133.00	118.19
hole2	8.90	127.10	118.20
hole2	24.50	142.41	117.91
hole3	8.10	126.24	118.14
hole3	10.90	128.60	117.70
hole3	13.27	131.01	117.74